

Universal Power Monitor - Body Setup and Calibration

([Click here for Summary Version](#))

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	30 mins	Not Applicable

2 Overview

This procedure sets up the Universal Power Monitors (UPM's) in Body Mode, by using the system and the Power Measurement Kit (Wattmeter). For proper calibration of the Universal Power Monitors, (UPM) both Forward and Reflected Power must be completed in Body mode.

Last Update	05/21/2007
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3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Power Measurement Kit (Wattmeter Kit)	1	-	2372868	-

3.2 Safety



⚠ DANGER

FATAL ELECTRIC SHOCK HAZARD!!

LIVE VOLTAGES THROUGHOUT THE CABINET WHEN ENERGIZED.

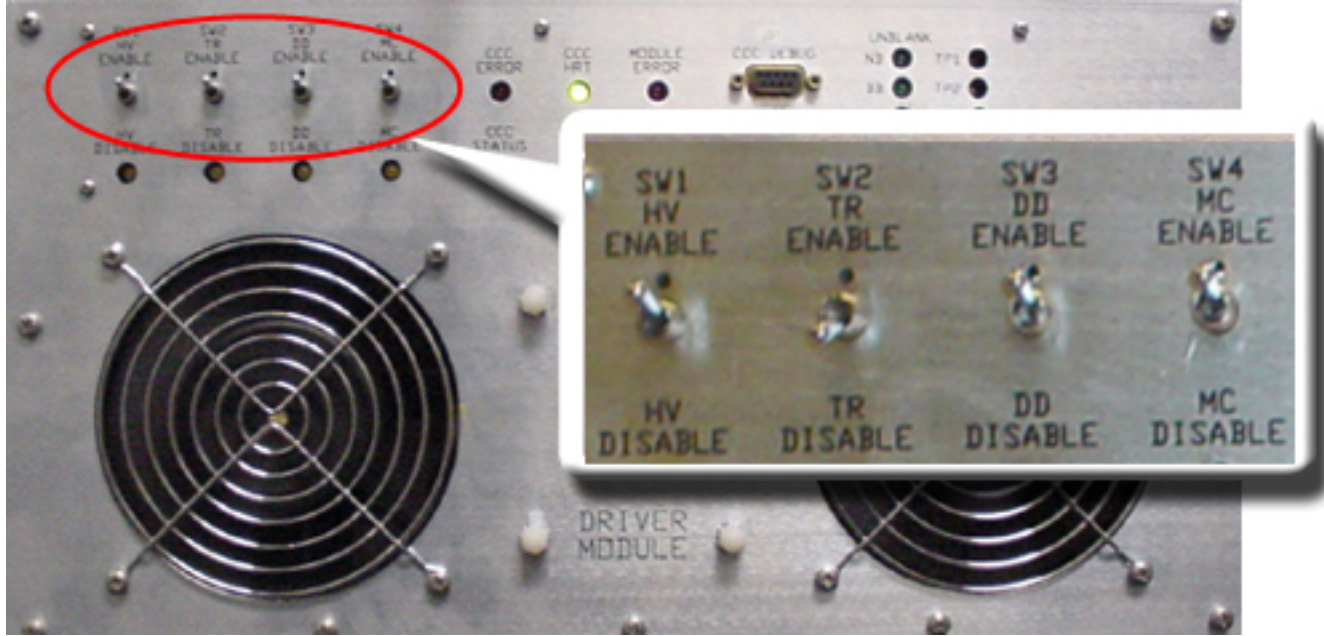
TO PREVENT FATAL ELECTRIC SHOCK, MAKE SURE THE RF AMPLIFIER IS DISABLED WHEN CONFIGURING THE CALIBRATION TOOLS.

4 Procedure

4.1 BODY FORWARD POWER CALIBRATION

1. Inhibit TR faults. Disable the Driver Module TR fault detection in the RFS (RF in the System Cabinet), (see [Illustration 1](#)). Move switch 2 to the TR Disable position.

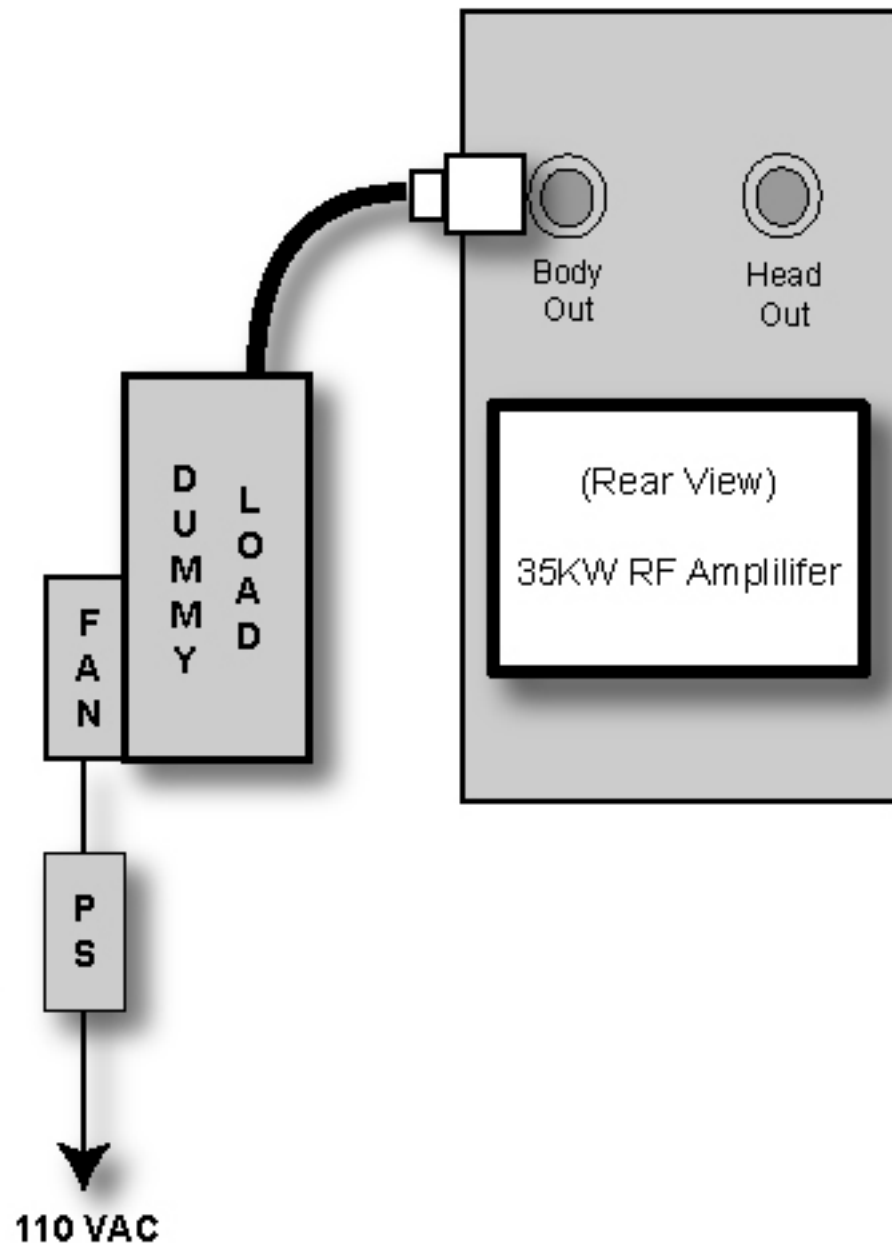
Illustration 1: Driver Module Switches



2. Remove the Body Heliax cable from J4. Connect the 35 KW RF dummy load from the Power Measurement Kit into J4. Plug the power supply on the dummy load into 110VAC service outlet in the System Cabinet. (see [Illustration 2](#)).

NOTE: If the RF Max power out procedure was just completed, configuration is the same. Leave the Body RF heliax cable disconnected and the head output can keep the coupler in line. It will not change the outcome of the calibration.

Illustration 2: Plug the Dummy Load into J4, Body Out



4.2 Setting up Software Tool to Calibrate UPM - Body Forward Power

This tool must see a Landmark only. It is not necessary to have a Body Phantom or Head coil in place, or any other scan parameter entered than what is described below. Simply Landmark on the cradle where the Head coil or Body Phantom would be.

1. Start a [New Exam]
2. Enter the following parameters:
 - a. **geservice**
 - b. Weight: **111**
3. Press the **Align** button on, (Land Mark).
4. Move the cradle to the Home position and then drive it into the bore.
5. Press **Landmark** button when the align light is approximately where the center of the Body Phantom would be. Press the **Advance to Scan** button.
6. From the Common Service Desktop Select:
 - a. Select [Calibration Tab]
 - b. Select [UPM Tool]
 - c. Start UPM tool from the Service Browser, (see [Illustration 3](#)).[Illustration 4](#) shows an example of the UPM tool.

Illustration 3: Starting UPM Tool

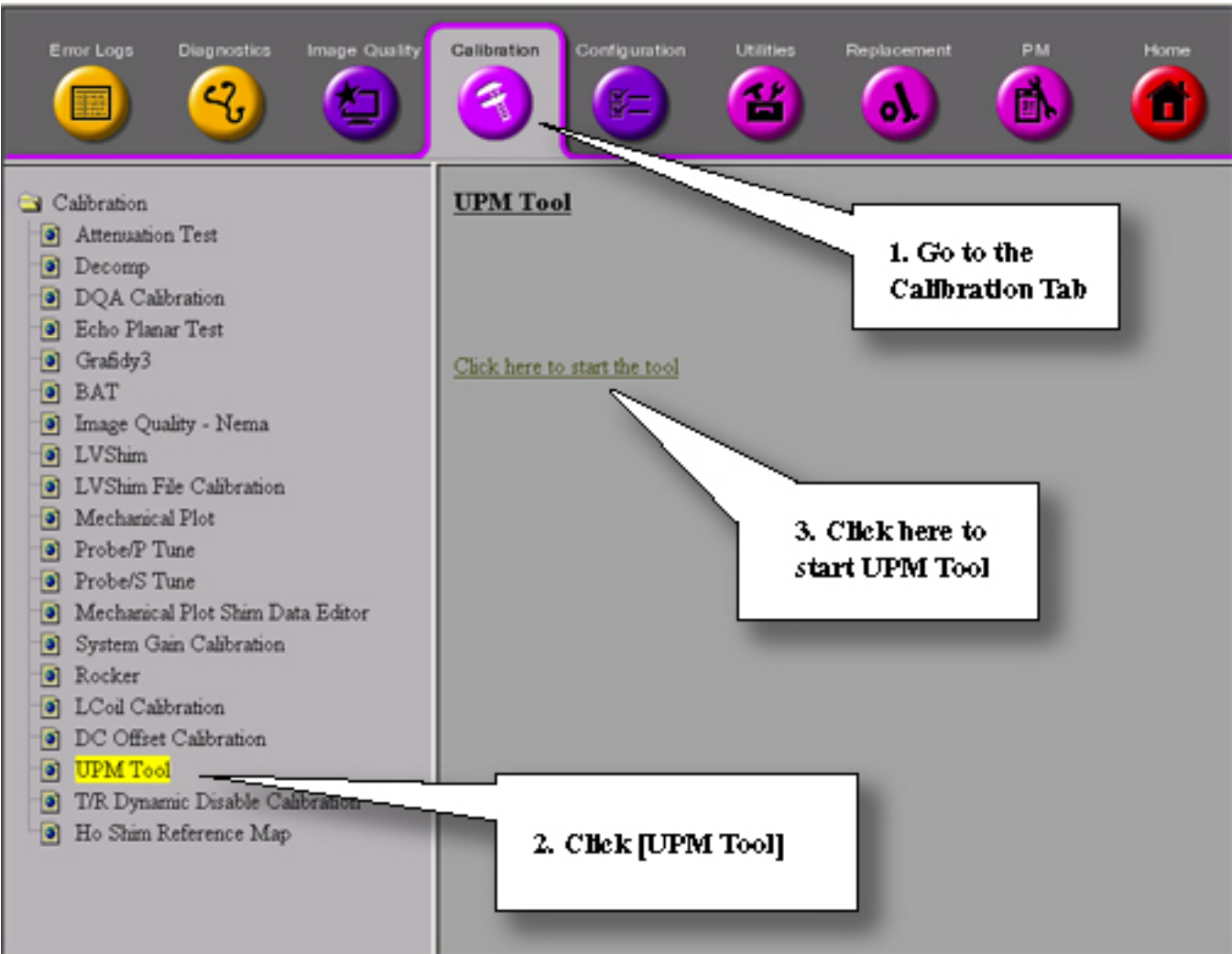
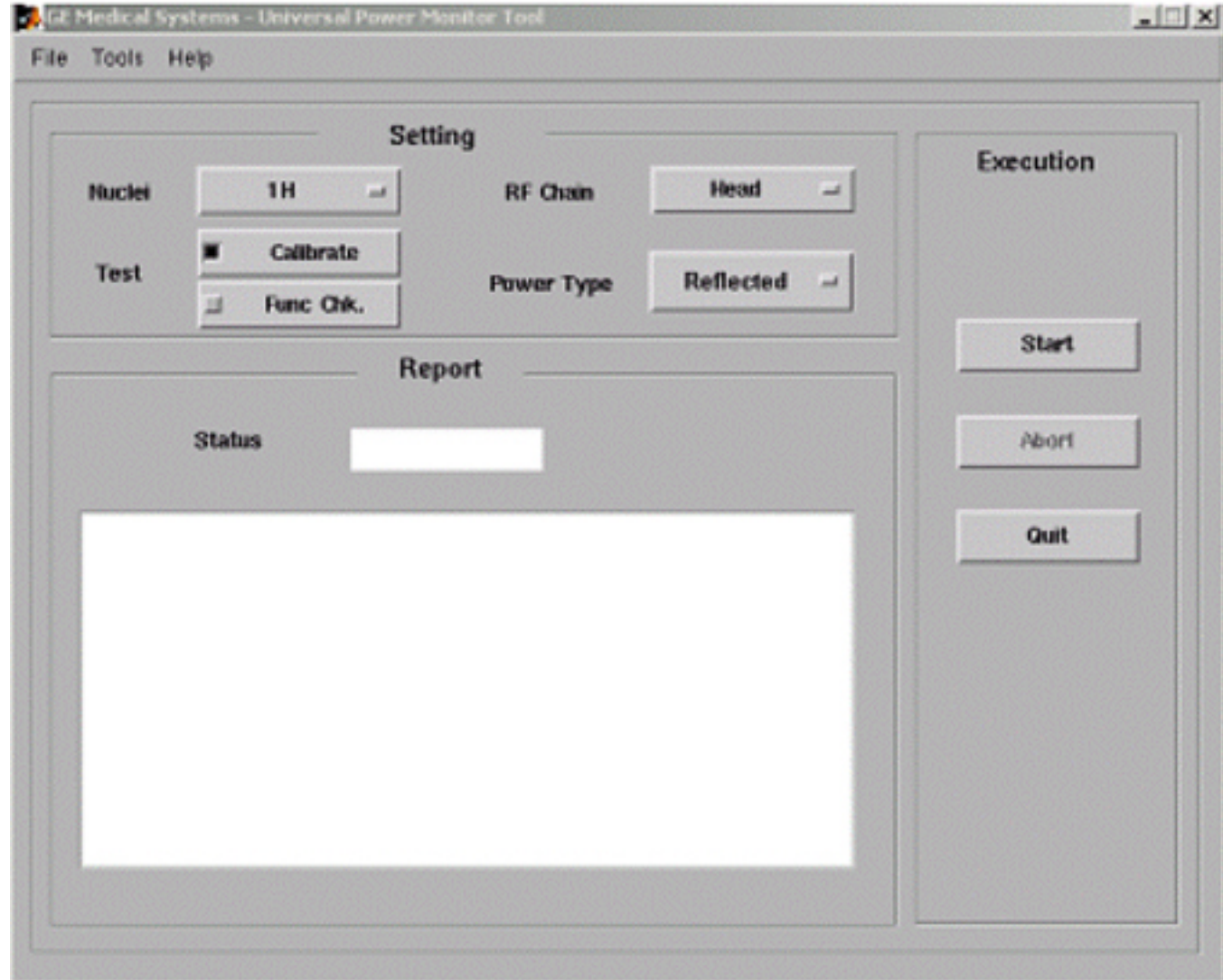


Illustration 4: UPM GUI



NOTE: The first time running the UPM Calibration tool after a load from cold, the following Error may appear: “The spectroRFampType and upm.config do not match! Please run the UPM Tool config file editor and hit [Restore Defaults] and save the config file”. If this error appears, click [File] -> “Edit Config”. When the window opens, click [Restore Defaults] then close the screen. The tool will now have the correct information and this pop-up will not appear again during future UPM Calibrations.

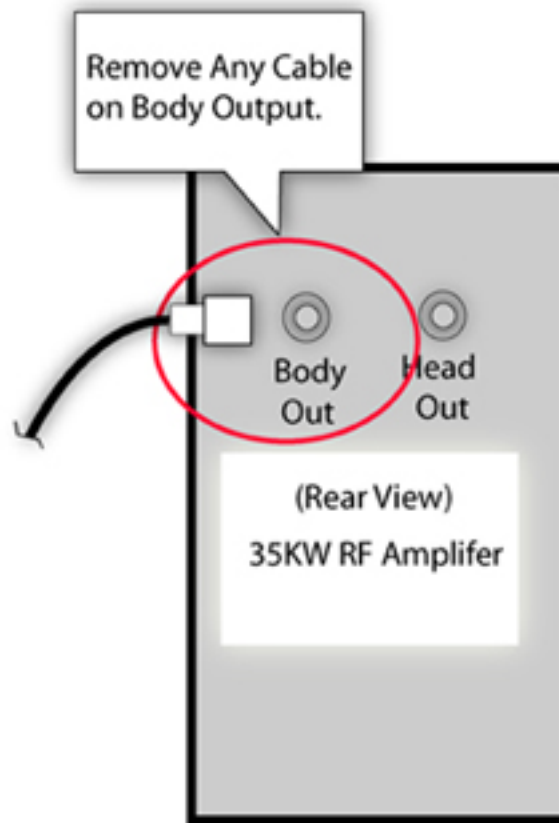
7. Select the Nuclei [1H].
8. Select the RF Chain you are going to calibrate [Body].
9. Select: [Calibrate] Button
10. Select Power Monitor type. [Forward]
11. Select the [Start] button on the tool opened earlier, (see [Illustration 4](#)).
12. Status should indicate Pass/Fail from 1 minute to 7.5 minutes depending on software version (tool should notify user of current step in progress). If failure occurs, see [Universal Power Monitor \(UPM\) Troubleshooting](#).
13. Proceed to the [UPM-Body and Head Functional Check](#) procedure, before moving to [Finalization, Step 1](#).

4.3 BODY REFLECTED POWER CALIBRATION

1. Disconnect the RF dummy Load From Body Output, J4. (see [Illustration 5](#)).

NOTE: Head cable may remain disconnected if the RF Body output calibration was just completed.

Illustration 5: Remove Cables from Output From J4



- If the amplifier is a 25 KW amplifier, disconnect the RF dummy Load From Body Output, J4. See [Illustration 6](#).

NOTE: If the amp at your site is a 25kW/35kW version with an LCD on the front, disconnect the Dummy Load from J4 directly as there are no RF couplers required for that amplifier. See [Illustration 7](#).

Illustration 6: Remove Cables from Output From J4 25 KW Amp

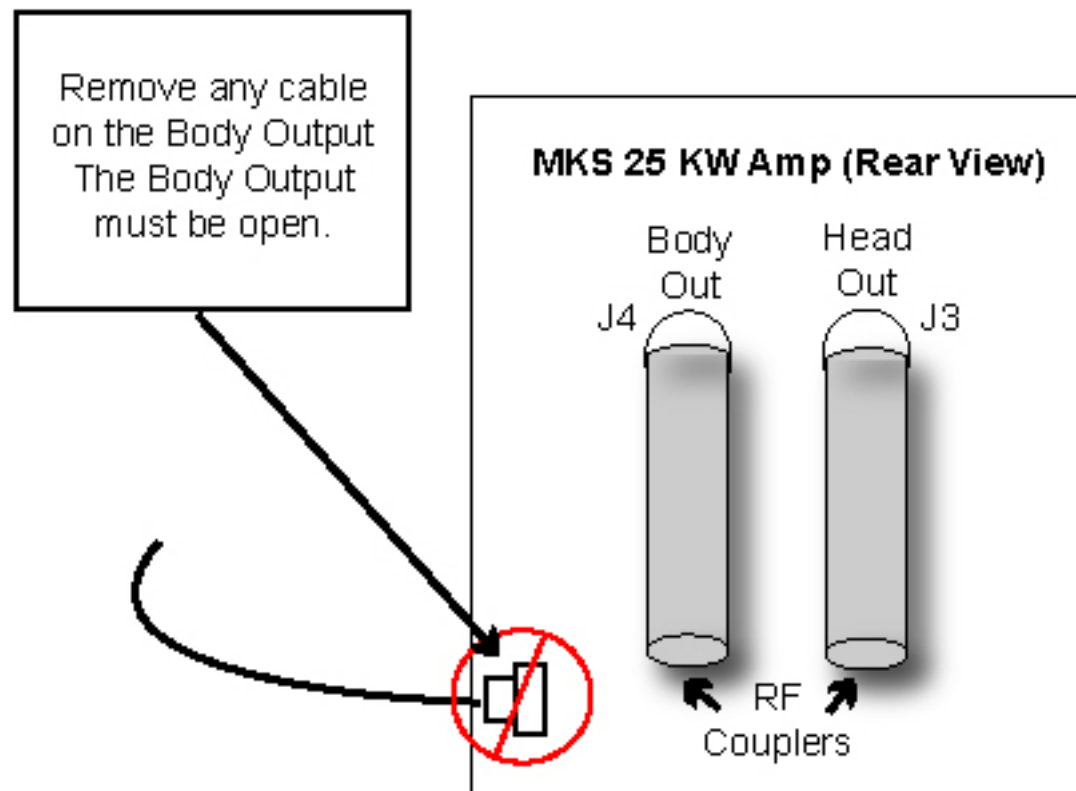
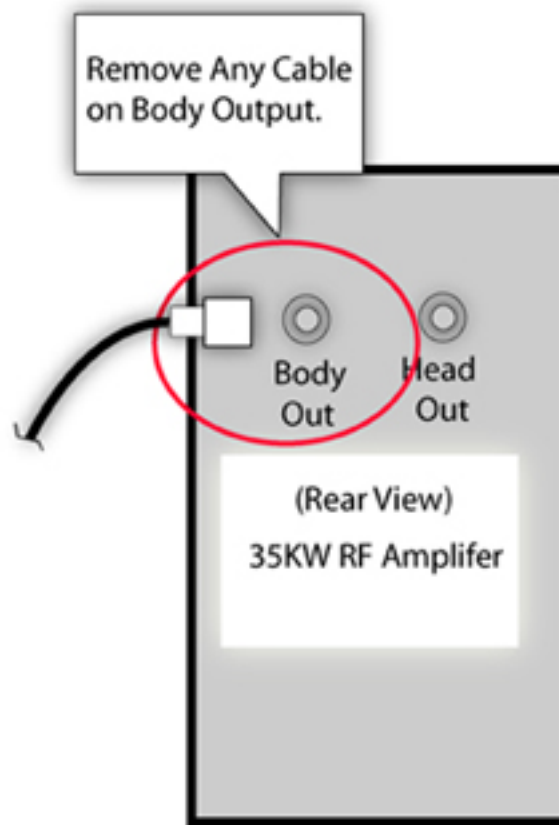


Illustration 7: Remove Cables from Output From J4 25kW/35kW KW Amp



3. Disconnect the RF dummy Load From Body Output, J4.
4. Locate and attach the cable stub in measurement kit to J4 on the 35 KW amp and the 25/35 KW amp. DO NOT attach the cable stud to the 25 KW amp.
5. Return to the UPM Tool User Interface. See [Illustration 4](#).
6. Make sure Nuclei [1H] is selected.
7. Make sure [Body] is selected.
8. Select Power Monitor type. [Reflected].
9. Make sure [Calibrate] is selected.
10. Select the [Start] button.
11. In 1 minute to 7.5 minutes depending on software version (tool should notify user of current step in progress and display Pass/Fail. If Failure occurs, see [Universal Power Monitor \(UPM\) Troubleshooting](#).
12. Proceed to the [UPM- Body and Head Functional Check](#) procedure, before going to [Finalization, Step 1](#).

5 Finalization

1. Remove the cables and power measurement equipment.
2. Re-attach the Body Helix Cables, or proceed to [Head RF Amp](#) calibration.
3. Re-enable T/R Drive fault detection. See Driver Module, [Illustration 1](#).
4. Upon completion of UPM calibration, [Reset TPS]. The system will not scan until this step is done